

Problems

Status

Standings

Fermat vs. Pythagoras

🔖 Tags:

Time Limit: 1 s

Memory Limit: 128 MB

Submission: 31

AC: 15

Score: 99.54

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Codes

AI Code Tips

Description

Computer generated and assisted proofs and verification occupy a small niche in the realm of Computer Science. The first proof of the four-color problem was completed with the assistance of a computer program and current efforts in verification have succeeded in verifying the translation of high-level code down to the chip level.

This problem deals with computing quantities relating to part of Fermat's Last Theorem: that there are no integer solutions of $x^n + y^n = z^n$ for $n > 2$.

Given a positive integer N , you are to write a program that computes two quantities regarding the solution of

$$x^2 + y^2 = z^2$$

where x , y , and z are constrained to be positive integers less than or equal to N . You are to compute the number of triples (x, y, z) such that $x < y < z$, and they are relatively prime, i.e., have no common divisor larger than 1. You are also to compute the number of values p such that p is not part of any triple (not just relatively prime triples).

Input

The input consists of a sequence of positive integers, one per line. Each integer in the input file will be less than or equal to 1,000,000. Input is terminated by end-of-file.

Output

For each integer N in the input file print two integers separated by a space. The first integer is the number of relatively prime triples (such that each component of the triple is $\leq N$). The second number is the number of positive integers $p \leq N$ that are not part of any triple whose components are all $\leq N$. There should be one output line for each input line.

Samples

input

Copy

```
10
25
100
```

output

Copy

1 4
4 9
16 27

Submit

Codes

[HZNUOJ](#) is based on [HUSTOJ](#)

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